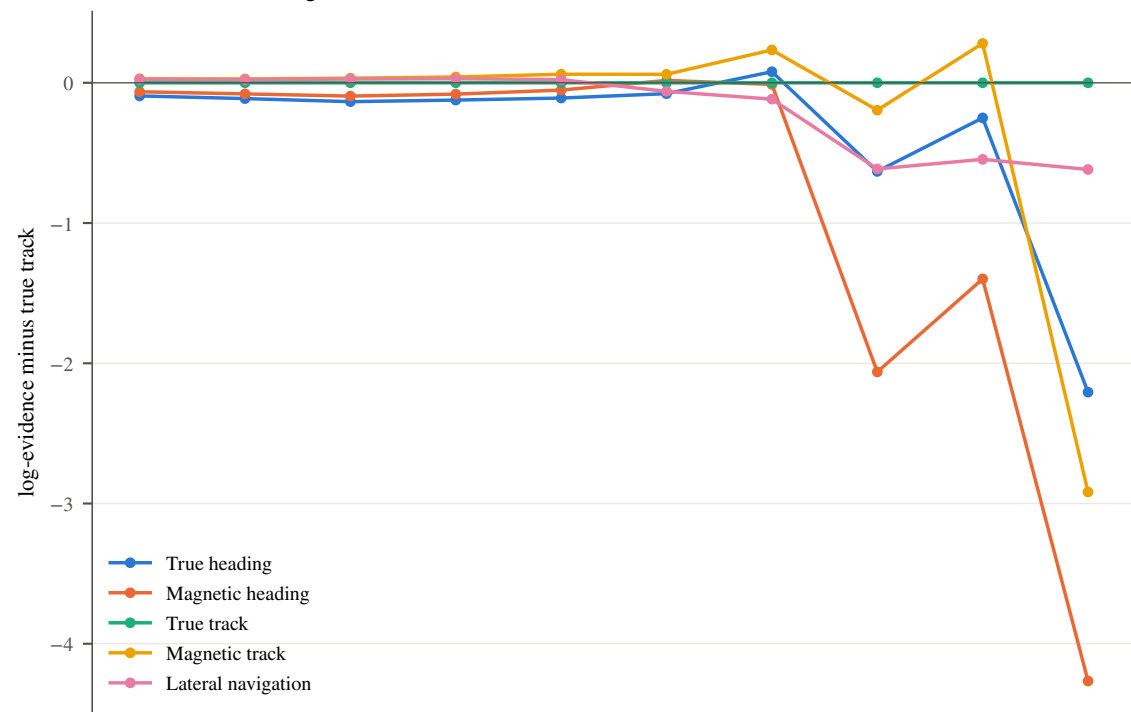
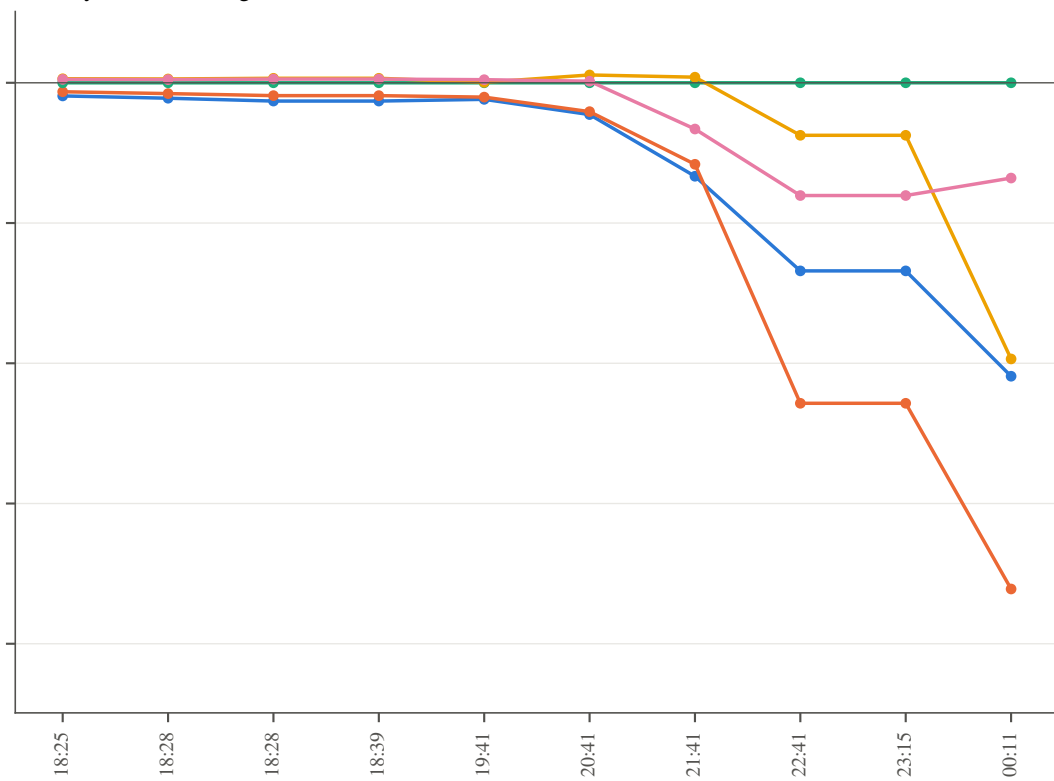


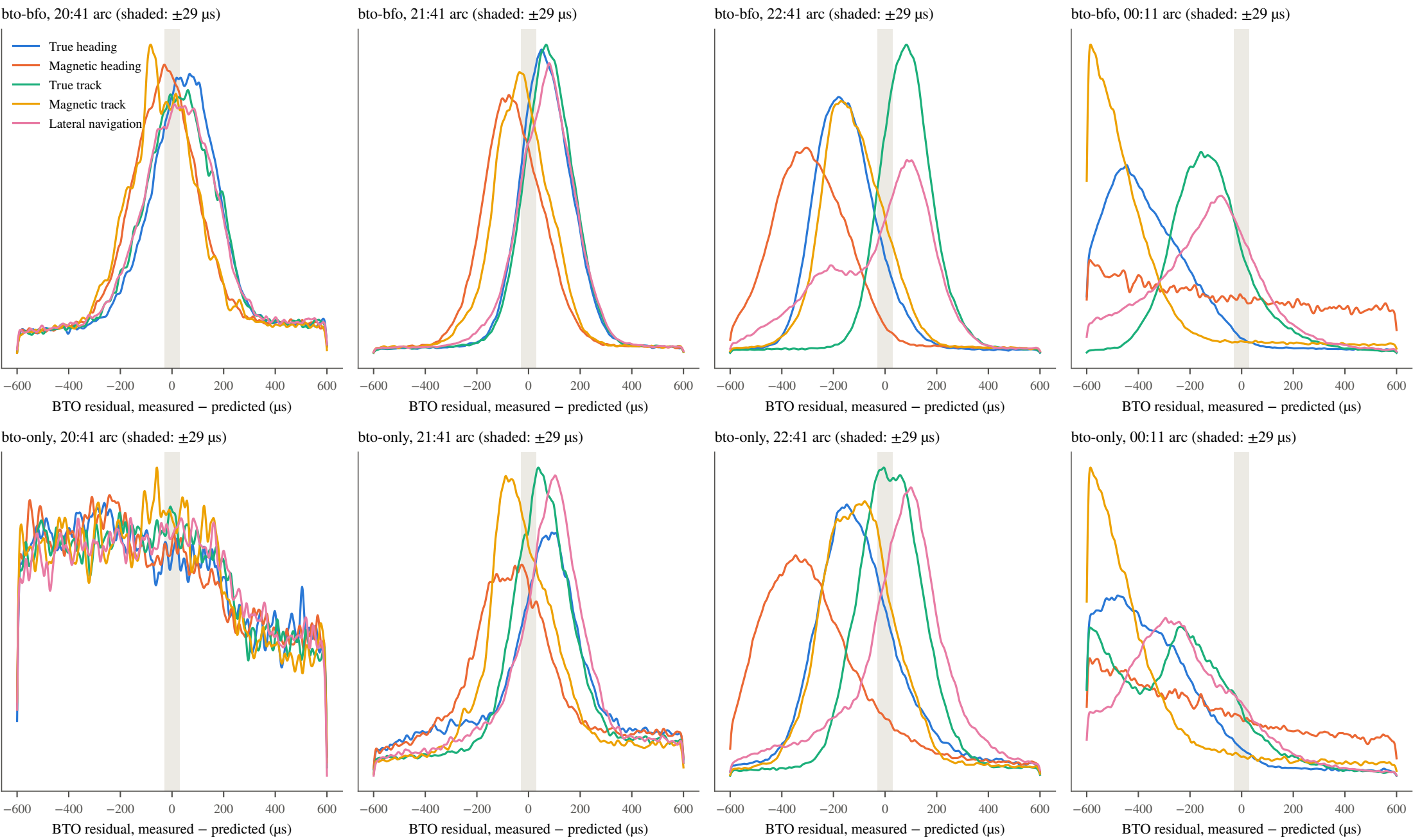
bto-bfo: cumulative log-evidence relative to true track



bto-only: cumulative log-evidence relative to true track

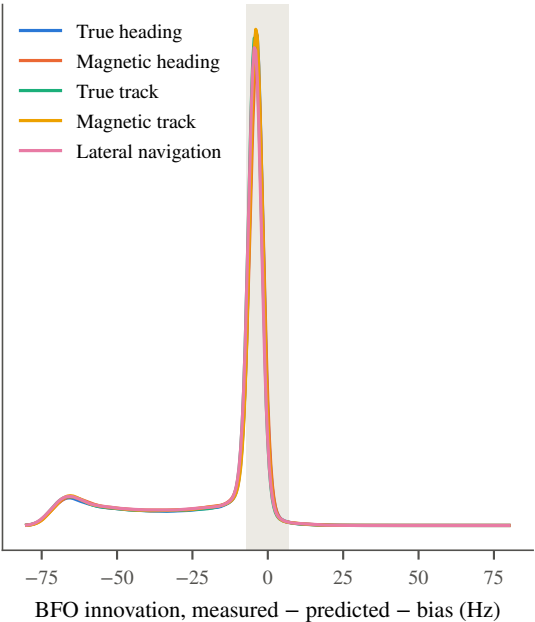


Negative values: the mode explains the measurements up to that arc less well than true track. Run residuals-to-0011; epochs excluded: m0019a, m0019b; replicates 1, 2; 100,000 predictive particles per mode and arc per replicate.

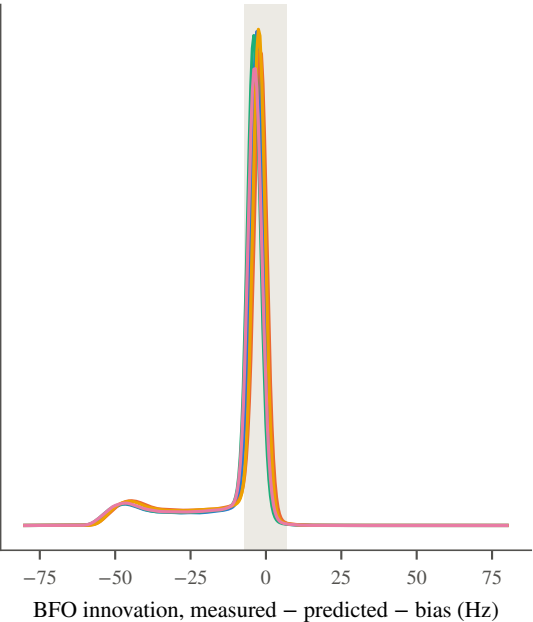


Predictive distribution of each mode's paths arriving at the arc (before its update). Positive: the path is still inside the arc (too close to the satellite). Run residuals-to-0011; epochs excluded: m0019a, m0019b; replicates 1, 2; 100,000 predictive particles per mode and arc per replicate.

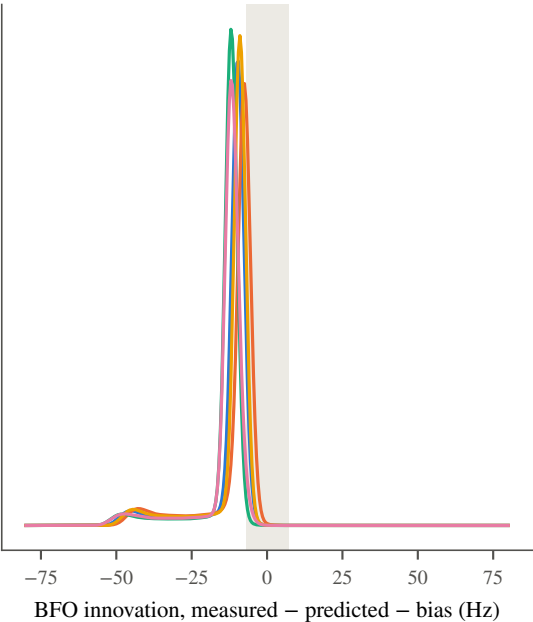
bto-bfo, 21:41 (shaded: ± 7 Hz)



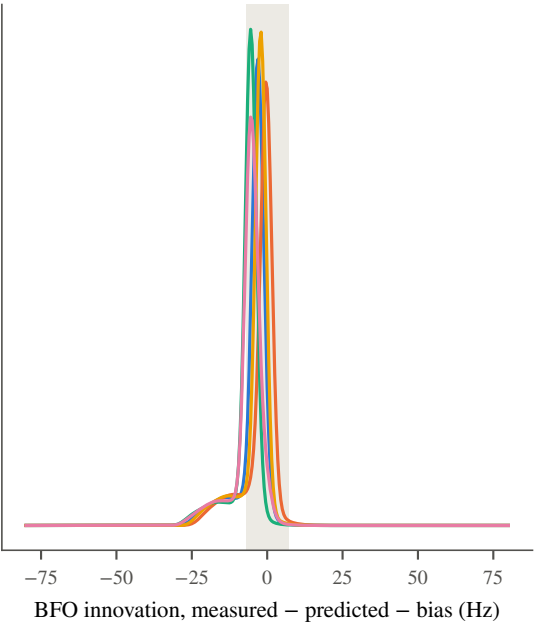
bto-bfo, 22:41 (shaded: ± 7 Hz)



bto-bfo, 23:15 (shaded: ± 7 Hz)

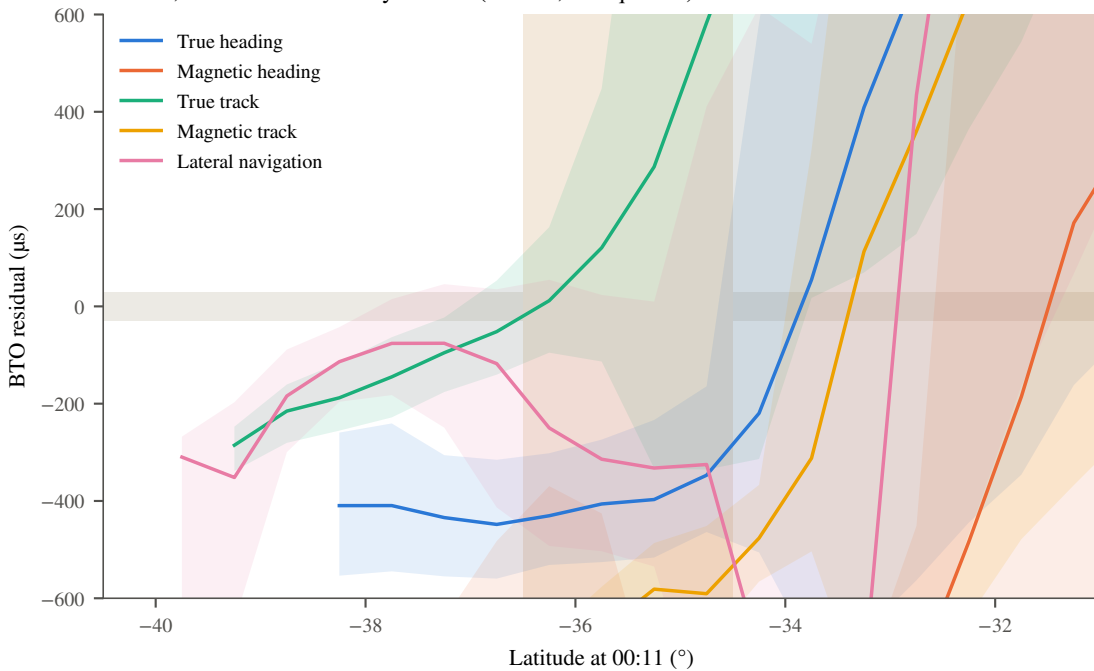


bto-bfo, 00:11 (shaded: ± 7 Hz)

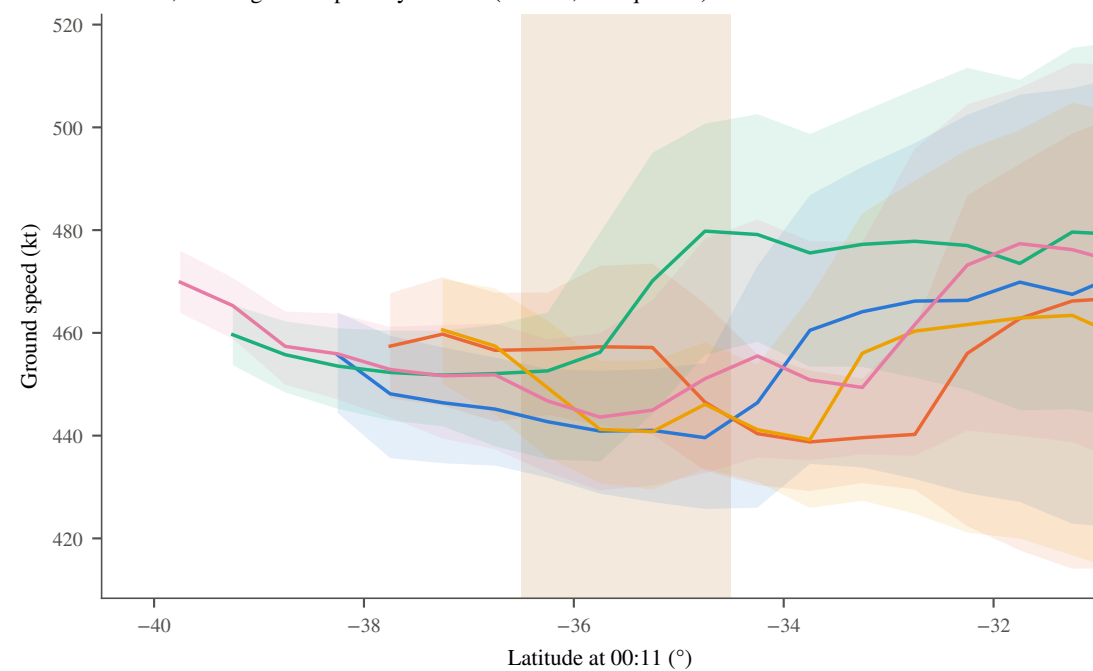


Innovation relative to each path's current bias estimate; its predictive s.d. also includes the bias uncertainty. Run residuals-to-0011; epochs excluded: m0019a, m0019b; replicates 1, 2; 100,000 predictive particles per mode and arc per replicate.

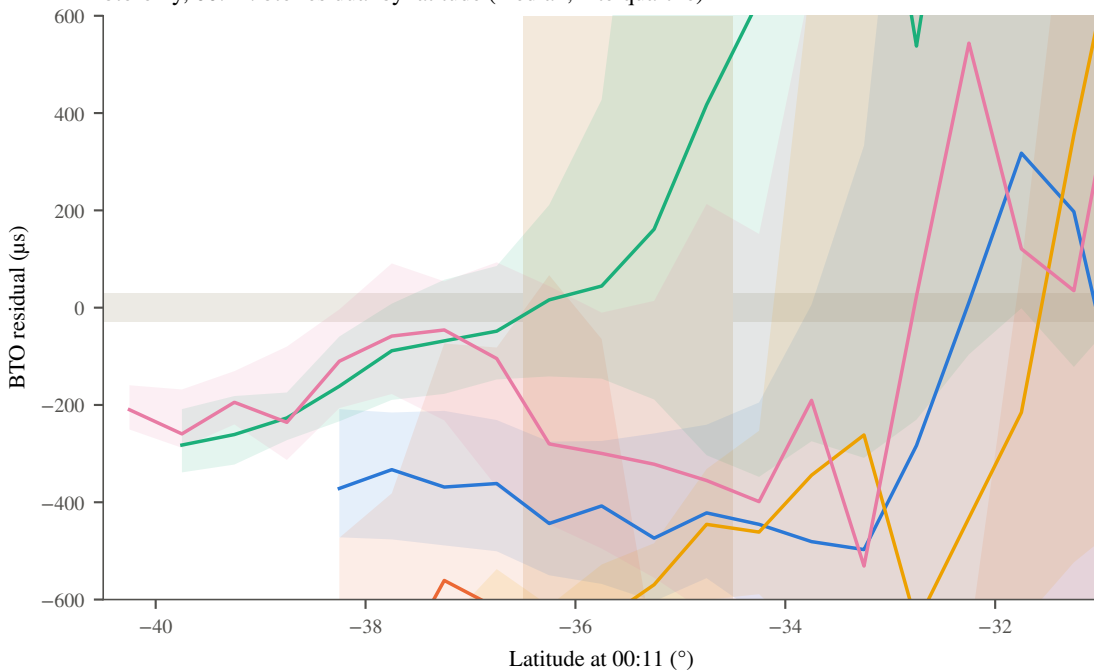
bto-bfo, 00:11: bto residual by latitude (median, interquartile)



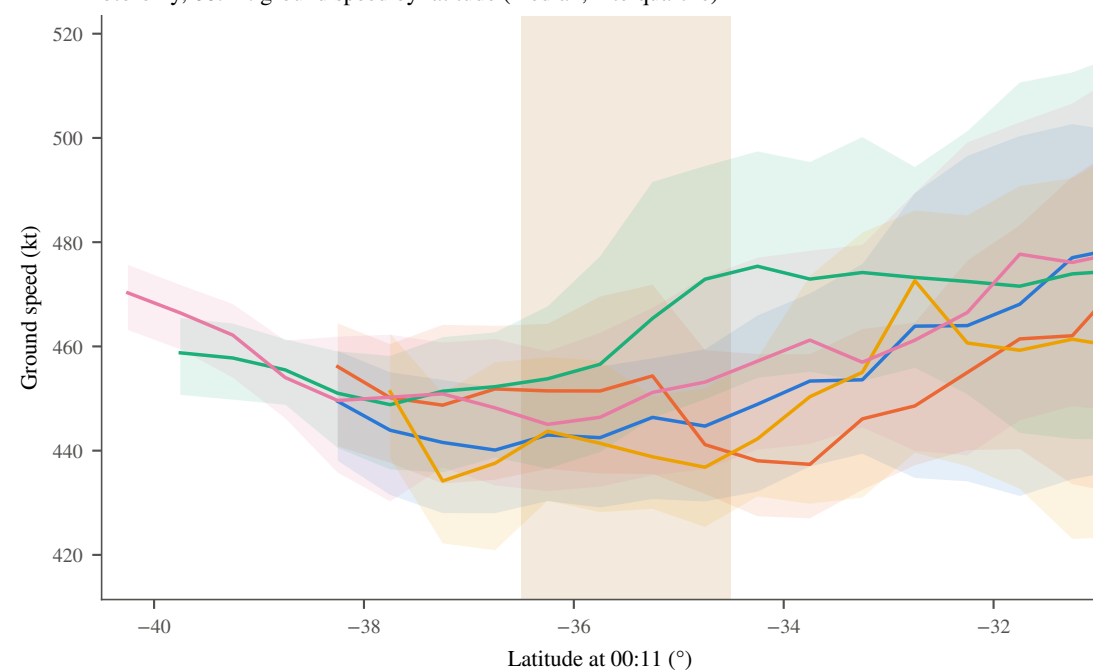
bto-bfo, 00:11: ground speed by latitude (median, interquartile)



bto-only, 00:11: bto residual by latitude (median, interquartile)



bto-only, 00:11: ground speed by latitude (median, interquartile)



Shaded vertical band: the published shoulder -34.5° to -36.5°. Particles from each mode's predictive distribution at 00:11; bins with fewer than 200 particles omitted. Run residuals-to-0011; epochs excluded: m0019a, m0019b; replicates 1, 2; 100,000 predictive particles per mode and arc per replicate.

Predictive residual summary by mode and arc

case	arc	mode	BTO mean (μs)	BTO s.d. (μs)	within $\pm 1\sigma$	BFO mean (Hz)	BFO s.d. (Hz)	ground speed (kt)	median lat
bto-bfo	20:41	True heading	+161	1173	6%	-26.6	28.7	470	-5.6
bto-bfo	20:41	Magnetic heading	+136	1172	6%	-26.9	28.8	469	-5.4
bto-bfo	20:41	True track	+138	1172	6%	-27.0	28.8	469	-5.5
bto-bfo	20:41	Magnetic track	+112	1173	6%	-26.6	28.7	469	-5.7
bto-bfo	20:41	Lateral navigation	+143	1180	5%	-27.5	28.9	469	-5.3
bto-bfo	21:41	True heading	+292	918	13%	-15.3	21.0	474	-16.7
bto-bfo	21:41	Magnetic heading	+209	946	11%	-15.6	21.2	471	-16.6
bto-bfo	21:41	True track	+296	915	12%	-16.0	21.2	473	-16.7
bto-bfo	21:41	Magnetic track	+221	936	14%	-15.8	21.2	469	-16.5
bto-bfo	21:41	Lateral navigation	+300	935	11%	-16.3	21.3	473	-16.6
bto-bfo	22:41	True heading	+130	970	6%	-9.1	13.7	479	-25.2
bto-bfo	22:41	Magnetic heading	+83	1065	2%	-9.0	14.4	469	-24.2
bto-bfo	22:41	True track	+334	943	13%	-10.1	13.6	475	-25.4
bto-bfo	22:41	Magnetic track	+201	1001	8%	-9.1	14.1	468	-24.5
bto-bfo	22:41	Lateral navigation	+248	983	8%	-10.0	13.9	476	-25.2
bto-bfo	00:11	True heading	-76	1091	1%	-4.8	4.9	446	-35.7
bto-bfo	00:11	Magnetic heading	-432	1286	0%	-2.4	5.2	448	-33.8
bto-bfo	00:11	True track	+128	1032	7%	-6.9	4.5	456	-37.4
bto-bfo	00:11	Magnetic track	-216	1189	0%	-4.1	5.0	448	-35.1
bto-bfo	00:11	Lateral navigation	+48	1099	7%	-6.4	4.9	455	-36.9
bto-only	20:41	True heading	-512	1713	1%	–	–	471	8.4
bto-only	20:41	Magnetic heading	-516	1717	1%	–	–	471	8.5
bto-only	20:41	True track	-475	1707	1%	–	–	469	8.3
bto-only	20:41	Magnetic track	-417	1689	1%	–	–	468	8.1
bto-only	20:41	Lateral navigation	-481	1710	1%	–	–	469	8.3
bto-only	21:41	True heading	+712	1484	3%	–	–	466	4.7
bto-only	21:41	Magnetic heading	+685	1492	4%	–	–	466	4.3
bto-only	21:41	True track	+683	1441	5%	–	–	462	7.4
bto-only	21:41	Magnetic track	+619	1421	5%	–	–	463	7.4
bto-only	21:41	Lateral navigation	+683	1446	4%	–	–	462	7.7
bto-only	22:41	True heading	+491	1332	6%	–	–	473	-23.0
bto-only	22:41	Magnetic heading	+407	1388	2%	–	–	468	-22.9
bto-only	22:41	True track	+437	1177	12%	–	–	458	19.7
bto-only	22:41	Magnetic track	+391	1230	8%	–	–	455	9.0
bto-only	22:41	Lateral navigation	+520	1268	7%	–	–	462	4.1
bto-only	00:11	True heading	+599	2168	1%	–	–	454	-34.6
bto-only	00:11	Magnetic heading	+932	2643	1%	–	–	457	-27.5
bto-only	00:11	True track	+372	1944	3%	–	–	454	38.8
bto-only	00:11	Magnetic track	+407	2163	1%	–	–	449	34.0
bto-only	00:11	Lateral navigation	+469	2035	3%	–	–	460	10.2

Run residuals-to-0011; epochs excluded: m0019a, m0019b; replicates 1, 2; 100,000 predictive particles per mode and arc per replicate.